

Machine Learning - Mid-Semester 1

Mock 01

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Machine Learning Midsem 1 - Mock 01

Time: 90 minutes

Maximum marks: 30

Instructions: Answer all questions. For calculations, show formula, substitutions, intermediate steps, conditions, shapes where relevant, and interpretation. No books or notes.

Q1 - Learning system design (Session 1, 4 marks, 10 minutes)

A factory wants to predict whether a pump will fail in the next seven days from sensor histories and known past outcomes.

- 1 Define task T , experience E , and one performance measure P . (2)
- 2 Name the learning paradigm and explain one way unrepresentative experience could invalidate the claimed performance. (2)

Q2 - Workflow and metrics (Session 2, 4 marks, 11 minutes)

A validation set yields $TP=21$, $FP=9$, $FN=7$, $TN=43$.

- 1 Compute precision and recall. (2)
- 2 The team scaled every feature using all records before splitting. Explain the problem and give the correct train/validation/test sequence. (2)

Q3 - Regression (Session 3, 4 marks, 13 minutes)

For a line with intercept, X has rows $(1,0), (1,1), (1,2)$ and $y=(1,3,5)^T$.

- 1 Distinguish the normal-equation system from its direct-inverse form: state both and the condition under which the inverse form is valid. (1)
- 2 Compute $X^T X$, $X^T y$, and solve for w . (3)

Q4 - Discriminants and cost (Session 4, 4 marks, 11 minutes)

A binary classifier has $g(x)=w^T x+b$, $w=(3,4)^T$, $b=-10$, and $x=(2,1)^T$. Disease posterior is 0.15; false treatment costs 2, missed disease costs 20, and correct decisions cost zero.

- 1 Define the decision boundary and compute score and signed geometric distance. (2)
- 2 Compute risks of treat/do-not-treat and choose the action. Explain why raw score, distance, posterior, and action are different objects. (2)

Q5 - Logistic and multiclass (Session 5, 4 marks, 12 minutes)

For three mutually exclusive classes the logits are $(\ln 2, 0, \ln 5)$ and the true class is class 1.

- 1 State the binary sigmoid/log-odds relationship and the scope of its convexity claim. (2)
- 2 Compute the softmax probabilities and one-example cross-entropy. (2)

Q6 - Decision tree (Session 6, 4 marks, 12 minutes)

A parent contains $(5+, 5-)$. A split creates equal-sized children $(4+, 1-)$ and $(1+, 4-)$.

- 1 Define entropy and compute the information gain in bits. (3)
- 2 State one valid policy for missing values and explain why a continuous feature need not be pre-binned. (1)

Q7 - Neighbour methods (Session 7, 3 marks, 10 minutes)

A query's three neighbours have labels A,B,B at distances 1,2,3.

- 1 Compute uniform and inverse-distance weighted predictions. (1)
- 2 Explain how kNN, locally weighted regression, and an RBF model use neighbours or distances differently. (2)

Q8 - Integrated evaluation (Session 8, 3 marks, 11 minutes)

A time-stamped fraud project repeatedly chooses features, hyperparameters, and a threshold using its newest month, then reports that month's score.

- 1 Contrast the team's repeated-newest-month workflow with a correct chronological train/validation/test protocol. Include both transformation fitting and what to do after a test period has influenced choices. (2)
- 2 Explain why threshold selection belongs before the one-time final test and name a metric aligned with missed fraud. (1)